

Designing and Evaluating a Community Fact-Checking System for WhatsApp

ANONYMOUS AUTHOR(S)

Crowdsourced fact-checking succeeds on public social media, but it remains unknown how these mechanisms function within the private, relational networks of encrypted chat apps like WhatsApp. We conducted a two-week in-situ field deployment of a custom fact-checking bot within a Colombian WhatsApp community. Participants (N=38) evaluated 40 items of organic viral media sourced directly from their own group chats. Mixed-methods analysis reveals that the dynamics of crowd wisdom fundamentally fracture in private spaces. Participants exhibited a pervasive default to truth, struggling significantly to reject false videos and protective warning chains. Moreover, politically homogeneous crowds merely amplified existing biases, and exposing users to the community's verdict drove blind conformity rather than factual calibration. We outline actionable design principles for encrypted moderation, highlighting the need for privacy-preserving cross-group aggregation, targeted modality friction, and the blinding of social feedback.

CCS Concepts: • **Human-centered computing** → **Empirical studies in collaborative and social computing**; *Collaborative and social computing systems and tools*.

Additional Key Words and Phrases: misinformation, WhatsApp, crowdsourced fact-checking, community notes, encrypted messaging, field study, Colombia

ACM Reference Format:

Anonymous Author(s). 2027. Designing and Evaluating a Community Fact-Checking System for WhatsApp. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '27)*. ACM, New York, NY, USA, 29 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

For billions of people, most of them in the Global South, WhatsApp is where news arrives, where family and neighborhood groups organize, and where information travels [28, 48, 63]. It is also a channel for misinformation with offline consequences, from public-health harms to lynchings triggered by kidnapping rumors [4, 6, 37, 42]. Because messages are end-to-end encrypted, the platform cannot rank, label, or remove content the way public platforms do. Professional fact-checking reaches WhatsApp mainly through tiplines that users must seek out, and these reach a small fraction of the content that circulates [34, 54]. Content moderation on WhatsApp falls almost entirely on community members themselves [19].

Relying on everyday users to police content is difficult in practice. Correcting misinformation inside group chats creates social friction. People hesitate to challenge friends, family, or community elders, and direct confrontation often damages relationships rather than correcting beliefs [19, 59, 64]. At the same time, individual users rarely verify what they receive. People generally default to accepting incoming information as true [10, 38], a bias that is amplified when content comes as realistic video [61, 67] or aligns with the recipient's political views [7, 23]. For human-computer interaction and social computing researchers, encrypted messaging represents the hardest test case for content moderation: how to design protective mechanisms for information ecosystems where the platform operator is intentionally blind to the content being shared.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2027 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

Manuscript submitted to ACM

The most developed alternative to centralized moderation is crowdsourced fact-checking, operationalized at scale in X's Community Notes [60, 68]. Its empirical basis is that small, politically balanced crowds of laypeople rate news about as well as professional fact-checkers do [2, 25, 44]. However, this empirical evidence may not transfer directly to a platform like WhatsApp, because: (1) participants share social ties rather than interacting as anonymous crowdworkers [3]; (2) viral content is rarely isolated text headlines, but rather local videos, images, and text chains; (3) communities are not artificially balanced, but reflect whatever demographic or ideological skew the local network possesses [9]; and (4) any deployed mechanism must expose social feedback, showing members what their peers concluded, which risks introducing conformity pressure and epistemic herding [39]. Whether a community's aggregated judgments can reliably identify false content in-situ, and how members behave when exposed to collective verdicts, has never been evaluated in the wild.

To address this gap, we built a WhatsApp-native verification system using the official Business Platform API and deployed it in a two-week field experiment in Colombia. We recruited 38 participants through local community networks. Crucially, rather than presenting synthetic lab stimuli, we used a data-donation protocol [20] to collect 807 viral items circulating inside participants' own WhatsApp groups, selecting 40 verified items (20 true, 20 false) spanning news, local alerts, and political commentary across video, image, and text formats. Each day, participants evaluated two incoming items as true or false and immediately received feedback showing the community's collective vote. Following the deployment, we conducted semi-structured interviews with nine participants across accuracy tiers.

Our findings reveal instances of where such a system works and where it fails. First, the deployment revealed a strong truth bias across the community. Participants reliably accepted true content but consistently struggled to reject false items. This vulnerability was highly concentrated in protective warning chains and false videos, which routinely fooled the majority. Partisanship further distorted judgments, as participants were far more likely to believe falsehoods that flattered their own side. Consequently, the accuracy of the crowd on political items depended on its ideological composition. In terms of aggregation, while simple majority voting proved effective overall, attempting to weight votes by past performance provided no real benefit. Individual accuracy fluctuated too much to reliably identify experts over a short deployment. Finally, showing participants the community's verdict did not calibrate them: learning that they had disagreed with the majority made them more likely to answer "true" on the next item, whichever way the community had voted, without improving their accuracy.

This work makes three contributions to the design and understanding of moderation in private spaces. Methodologically, we contribute an ecologically valid deployment architecture for studying encrypted networks, testing a fact-checking intervention directly within WhatsApp using organic media from participants' own groups. Empirically, we provide a behavioral account of peer verification in the wild, revealing how collective sensemaking is derailed by modality effects (specifically video), ideological homophily, and the effect of peer feedback on later judgments. Finally, we translate these findings into design principles for decentralized moderation, outlining how platforms can structure cross-group aggregation, delay reputation weighting, and present peer verdicts without breaking end-to-end encryption.

2 Related Work

To situate our field deployment, we draw on three distinct bodies of literature. We first review the mechanics and limitations of crowdsourced fact-checking on public platforms. We then examine the cognitive heuristics and biases that drive how individuals evaluate misinformation in practice. Finally, we explore why the architectural and relational dynamics of WhatsApp complicate these behaviors, creating unique barriers for community-driven moderation.

2.1 Crowdsourced Fact-Checking and Its Boundary Conditions

The premise of crowdsourced fact-checking relies on classic aggregation theory: if laypeople are even slightly better than chance at identifying truth, and their errors are independent, their collective judgment will converge on accuracy [26, 49, 62]. On public platforms, this dynamic largely holds. Aggregated lay judgments correlate with professional fact-checkers just as well as professionals correlate with each other [2, 25, 44], and crowds can reliably flag unreliable sources to prioritize verification pipelines [9, 17, 36].

However, the empirical success of systems like X's Community Notes [13, 15, 60, 68] masks vulnerabilities that become critical in smaller, private networks. Fact-checking is highly partisan: contributors routinely penalize content from political out-groups while giving a pass to their own [3, 14, 53, 57]. Community Notes neutralizes this using bridging algorithms that publish notes only when raters across deep ideological divides agree [68]. But a typical WhatsApp community lacks the massive, diverse rater pool required to make bridging work. Furthermore, aggregation theories warn that when raters can see each other's judgments (as they inevitably would in a localized chat environment) the independence required for crowd wisdom collapses, inflating confidence without improving accuracy [32, 39]. While some factual estimation tasks can survive social influence [8], partisan priming typically destroys this benefit entirely [27]. Whether social feedback acts as a helpful signal or a pull toward blind conformity in encrypted chats remains an open question, which we test directly.

2.2 How People Judge Misinformation, and What Interventions Change

When encountering dubious claims online, people rarely pause to verify them. Instead, they rely on fast heuristics like source reputation, social endorsement, and production quality [18, 47]. Because social feeds are optimized for passive consumption, users operate with a default bias toward belief, accepting incoming statements unless a specific cue triggers suspicion [10, 24, 38].

Signal detection studies separate this baseline response bias (the criterion) from the actual ability to discern truth (sensitivity). They consistently find that ideological alignment primarily shifts the criterion: people simply lower their threshold of skepticism for claims that favor their political side [7, 23, 51, 52]. Modality further short-circuits skepticism. Video is persistently judged more credible than text, as the visceral realism of a clip substitutes for missing provenance cues [29, 61, 67].

The HCI community has developed various interventions to counter these defaults, from interfaces that scaffold lateral reading [58] to reflective prompts that disrupt automatic sharing habits [16, 31, 35, 66]. Yet, these interventions are almost exclusively designed for and evaluated on public social media feeds or in laboratory surveys. We extend this work by deploying an intervention entirely inside WhatsApp, combining the social feedback of peer moderation with ecologically valid multimedia content.

2.3 WhatsApp as a Context for Community Moderation

WhatsApp presents a fundamentally different moderation environment than public networks. First, end-to-end encryption forces researchers and fact-checkers into the dark. Without a platform-level view, studies rely on data donated by consenting users or scraped from public invite links [20, 22, 55]. This research reveals that viral WhatsApp content is overwhelmingly visual and intensely local [21, 22], creating a massive blind spot for centralized fact-checking organizations, which rely on underutilized tiplines to discover rumors [34, 46, 54].

Second, moderation within WhatsApp groups is relational, not rule-based. Group administrators derive their authority from real-world offline standing and prioritize social harmony over policing conversation [59]. As a result, explicit correction in group chats is exceptionally rare [19]; users fear social friction, desire neutrality, and severely doubt that deliberation can succeed in politically homogeneous circles [1, 64].

Third, when users do attempt to correct misinformation, the success of the intervention depends almost entirely on the social ties between the speaker and the recipient. Correcting a peer is not merely an exchange of facts; it is a relational act. Research consistently shows that debunking efforts are far more persuasive when they originate from strong, in-group ties rather than anonymous strangers or institutional fact-checkers [50]. This dynamic spans across platforms, even on public networks like Twitter, users are significantly more likely to accept a correction if it comes from a mutual connection rather than an unknown account [43]. Because WhatsApp is anchored in real-world offline networks, these social requirements for trust are especially critical.

However, relying on these social ties for moderation introduces its own set of frictions. Social corrections on WhatsApp rarely occur in an objective vacuum; instead, they frequently bleed into heated political arguments and cross-cutting ideological exposure [56]. Furthermore, the ability to correct someone is heavily constrained by cultural and familial hierarchies [41]. In many communities, publicly challenging an elder, an uncle, or a respected group administrator who shares a false rumor violates social etiquette, effectively stifling peer moderation. These demographic and hierarchical tensions are vital to understand, as age dictates much of how misinformation travels on the platform. Older users, for instance, often circulate alarmist warnings or partly true content out of a genuine desire to protect their families, and they process and react to corrections differently than younger digital natives [65].

Synthesizing these three domains reveals a critical gap: while we understand how crowds aggregate on public feeds and how individuals succumb to cognitive biases in the lab, we do not know how these forces operate on private, and encrypted chat apps. This study bridges that gap. By deploying a fact-checking bot directly within a WhatsApp community and testing it exclusively with their own viral media, we move crowdsourced moderation out of artificial laboratory settings. This allows us to observe how community verification actually functions in the wild, providing the first empirical measure of how lay groups evaluate encrypted media in situ.

3 System Design and Field Methodology

We conducted a two-week field experiment using a mixed-methods design: a quantitative phase where participants judged viral content daily via a custom WhatsApp bot, followed by semi-structured qualitative interviews. The study was approved by our institutional review board.

3.1 Setting and Participants

The study took place in Colombia, where WhatsApp is the main channel for personal, work, and civic communication and a primary source of news [48]. We recruited participants between July 21 and August 1, 2025 by snowball sampling that began in the personal network of the first author, who is from the community. Recruitment through known contacts was a deliberate choice, as the interviews later confirmed, unsolicited messages from unknown numbers are widely assumed to be scams, and a vouching contact was the condition for participation. Each participant joined through a short video call in which the study was explained and informed consent obtained.

Forty people received items; two never answered and are excluded, leaving 38 participants. Residence was recorded at registration for 27 of them: 13 lived in Candelaria, a municipality in the Valle del Cauca department in the southwest, 13 in the Caribbean departments of Cesar, Magdalena, and Atlántico, and 1 in Caldas. Thirty-six participants completed

the pre-survey. Of these, 21 were women and 15 men; ages ranged from 18 to 64 (mean 33.6, SD 15.1), and years of formal education from 2 to 24 (mean 12.9, SD 4.7). When asked which group of national politicians they identified with, 17 chose the Left, 14 the Right, and 5 the Center. The sample was therefore close to balanced between Left and Right, a point we return to in Section 4.3. On a three-item cognitive reflection test, 6 participants answered none correctly, 8 one, 9 two, and 13 all three. Twenty-eight reported using the internet and social media at least “often” (4 or 5 on a 5-point scale). Appendix A (Table A.1) lists every participant.

Participants received 20,000 Colombian pesos (about USD 5) on registration, and 40,000 or 50,000 pesos (about USD 10 or 12.50) at the end of the study if they had responded to at least 60% or 90% of the items, respectively.

3.2 Content Collection and Stimulus Selection

To ensure high ecological validity, we sourced our stimuli directly from the participants’ own chats. After consenting, participants used WhatsApp Explorer [20], a data-donation tool that allows users to selectively export group chat data to donate media from their communities. By filtering exclusively for messages flagged by WhatsApp as “forwarded many times,” we collected 807 highly viral items circulating across 243 groups during July and August 2025.

The first author (a native Spanish speaker from the community) manually screened this corpus to isolate verifiable factual claims, filtering out jokes, greetings, opinion pieces, and ambiguous satire. We then established ground truth for the remaining items: false items had to be explicitly debunked by a professional fact-checking organization (such as Colombiacheck or AFP Factual Colombia), while true items had to be directly traceable to a primary source, such as official government announcements, legislative text, or mainstream news reports.

From this verified pool, we selected a final stimulus set of 40 items (20 true, 20 false) that reflected the actual modalities and topics circulating in the community.¹ Prior to deployment, both authors independently cross-verified all 40 labels against their primary sources. The final set was heavily multimodal, comprising 18 images, 15 videos, 5 text messages, and 2 text-with-image messages.

We coded each item by genre and, where applicable, partisan slant, using the coding made when the items were selected: an item favors a side if its central claim attacks a named politician or party of the other side or defends one of its own. Two false political items whose target was ambiguous, a montage laying a foreign president’s speech over images of the Colombian right and a post conflating several corruption cases, were coded as having no identifiable slant; Appendix F shows that the partisan results hold when they are coded as favoring the Left. The 20 false items included 8 pieces of political fabrication (e.g., doctored charts, manipulated quotes), 8 protective “warning” chains (e.g., kidnapping alerts, scam warnings, fake legal advice), 3 health rumors, and 1 conspiracy infographic. The 20 true items included 12 news reports, 6 informative or nature items, and 2 health recommendations. Notably, the partisan slant of the false items was asymmetric: five favored the Left and one favored the Right, while one true item favored the Right. This imbalance was not engineered and it simply reflects the organic distribution of political media circulating in the participants’ groups at the time, a dynamic we account for in our partisan analyses. Appendix B (Table B.1) details all 40 items.

¹Seventeen of the false items are fabrications or hoaxes. Three are misleading rather than fabricated: a list of the president’s “broken promises” that mixes real and false entries, and two propaganda videos that recontextualize real speech (LC26, LC29, and LC39 in Table B.1). Appendix F reports the main results without them.

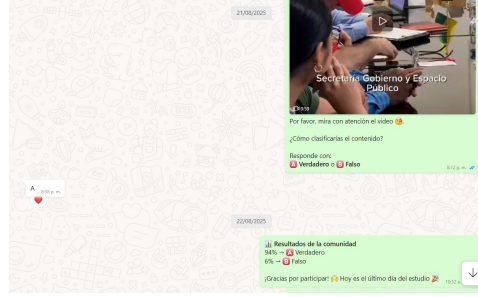


Fig. 1. A participant’s view of the bot. A video arrives with the prompt “Please watch the video carefully. How would you classify the content? Reply with A True or B False”; after the reply, a message headed “Community results” reports the percentage of participants who answered true and false.

3.3 System Architecture and Privacy

We built the fact-checking bot using the official WhatsApp Business Platform API, hosted on a cloud server. While the API guarantees reliable media delivery and supports interactive responses, it enforces strict anti-spam constraints: a business account can only message a user freely within a 24-hour window following that user’s last message. We designed the daily protocol around this architectural constraint, ensuring that each participant’s reply kept the communication window open for the subsequent item. Participants submitted their judgments simply by replying with a corresponding letter (see Figure 1).

We prioritized participant privacy throughout the data pipeline. All data were pseudonymized at collection, with phone numbers replaced by random identifiers. The mapping between phone numbers and identifiers was stored in a separate, encrypted file accessed solely for the purpose of paying incentives; the bot’s active response logs contained only the pseudonymous IDs. Furthermore, the bot only ingested content explicitly marked by WhatsApp as “forwarded many times” from the donated groups. Finally, to ensure participants were not left misinformed by the study itself, the debriefing at the end of the deployment explicitly revealed which of the 40 items had been true and which had been false.

3.4 Deployment Procedure

Figure 2 summarizes the three phases of the study. The quantitative deployment ran on weekdays from August 11 to 22, 2025. We assigned participants at random to one of four fixed schedules, created by crossing two distinct item sets in week one with two item sets in week two. The four schedules were completed by 8, 11, 11, and 8 participants, respectively. Each schedule comprised 20 items (ten per week) so that every participant judged exactly half of the 40-item pool, and each item was sent to 19 participants. We capped the workload at 20 items per participant (rather than the full 40) to keep the daily burden at just two items and minimize attrition. Within each schedule, the order of items was fixed and identical for all participants assigned to it, and each item appeared at the exact same position in both schedules that contained it. Because position and item are completely confounded in this design, we cannot isolate longitudinal learning or fatigue effects, a limitation we address in Section 4.4.

Each morning and afternoon, the bot sent one item with the following prompt: “Please look carefully at the following video/image/text. How would you classify the content? A – True. B – False.” After the participant replied, the bot

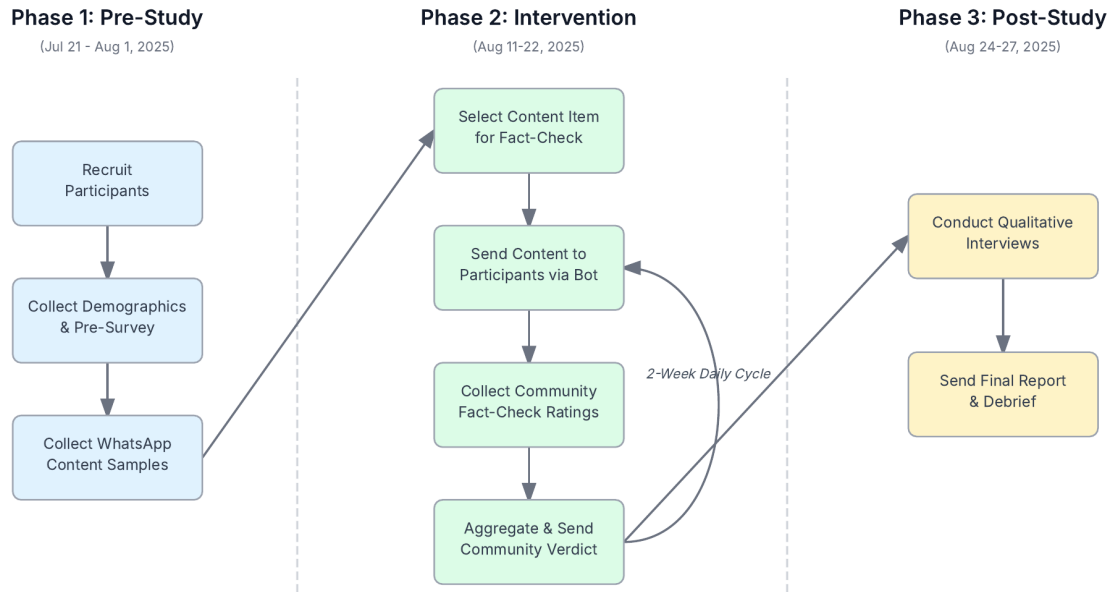


Fig. 2. Study protocol. Phase 1 (July 21 to August 1, 2025): recruitment, pre-survey, and content donation. Phase 2 (August 11 to 22): a daily cycle in which the bot sent an item, collected the participant’s true/false judgment, and returned the community’s vote. Phase 3 (August 24 to September 10): interviews and debriefing.

returned the community verdict for that item. For example: “Here is how your community responded to this message: 94% → True, 6% → False. Thank you for participating!”

The percentages sent for an item equaled its final tally: the verdict was compiled once the item’s responses had been collected and delivered before the next item, so every participant saw the same verdict for a given item, and participants who had not answered an item also received its verdict. We integrated this social feedback for two reasons: first, any deployed crowdsourced system would inherently show participants the aggregate judgment of their peers; second, we hypothesized that seeing the community’s consensus might help less accurate participants calibrate their judgments over time. Crucially, because the verdict was revealed only *after* a participant submitted their answer, it could not influence their judgment on that specific item, allowing us to strictly isolate its effect on subsequent judgments (analyzed in Section 4.5). If participants fell behind schedule, the first author sent them a personal reminder to complete their pending items.

Engagement was high: participants answered 662 of the 760 items sent (an 87% response rate). The median participant completed 19 of the 20 items (mean 17.4, range 9 to 20). Overall, 27 of the 38 participants completed at least 90% of their items, and 36 completed at least 60%. Each of the 40 items ultimately received between 14 and 19 independent judgments (mean 16.6).

3.5 Post-Study Interviews

Following the quantitative deployment, we invited participants from each accuracy tercile to participate in a 20- to 30-minute semi-structured interview conducted in Spanish over Zoom by the first author between August 28 and

September 10, 2025. Nine participants took part. This included three individuals with high accuracy (above 0.7: P01, P08, P09; scoring 0.80, 0.78, 0.73 respectively), three with middle accuracy (between 0.65 and 0.7: P02, P04, P07; scoring 0.68, 0.69, 0.70), and three with low accuracy (at or below 0.53: P03, P05, P06; scoring 0.53, 0.53, 0.36).

The interviews explored four main areas: (i) the participant’s motivations for joining the study; (ii) their cognitive process for deciding whether an item was true; (iii) their emotional and behavioral reactions to seeing the community verdict; and, (iv) their feedback on the bot, specifically regarding what a non-compensated, real-world version of the tool would require to succeed. The interviews were recorded and transcribed. Participants designated P01 through P09 in Appendix A (Table A.1) are these interviewees.

3.6 Analysis Strategy

Units of analysis. Our data span three natural units: 662 judgments, 38 participants, and 40 items. Because judgments are inherently clustered within both participants and items, and our stimulus set contains only 40 items, we report primary statistical tests at the participant and item levels. We treat judgment-level models strictly as supporting evidence. To estimate these judgment-level models, we used generalized estimating equations (GEE) with an exchangeable correlation structure clustered on the participant. As a robustness check, we also fit logistic regressions with crossed random intercepts for participant and item [5, 33]. Because a maximum-likelihood mixed-model implementation was unavailable to us in Python, we fit these crossed models using variational Bayes in statsmodels. Both sets of estimates, together with the same GEE models clustered on items instead of participants, are reported in Appendix D (Table D.1).

Individual judgments. For each participant, we computed the hit rate (true items judged true) and false-alarm rate (false items judged true). From these, we calculated the signal detection measures d' (sensitivity) and c (criterion), applying the standard log-linear correction to account for rates of 0 or 1 [30, 40]. A negative c indicates a response bias toward answering “true.” We compared accuracy between true and false items within participants using a paired t -test and a Wilcoxon signed-rank test, and across items using Welch’s t -test.

Item characteristics. We analyzed accuracy across modality, genre, and partisan slant at the item level, and via judgment-level GEE models that included interaction terms for veracity. To assess partisan congruence, we isolated the six false items with a clearly identifiable political slant. We evaluated these specific items against the 31 participants who explicitly identified with the Left or the Right. A judgment was coded as “congruent” if the item’s slant favored the participant’s own political side. Appendix F (Table F.1) repeats the item-level analyses without the three false items whose labels are debatable and with the alternative slant coding described in Section 3.2.

Aggregation. To model crowd performance, we simulated crowds ranging in size from 1 to 38 members by sampling participants without replacement (generating 3,000 crowds per size). For each item, we computed the majority verdict from the simulated crowd members who had actually seen it, scoring exact ties as half correct. We evaluated four distinct weighting rules: unweighted voting; votes weighted by the participant’s accuracy across all their items (a naive approach that leaks information by using the target item to estimate the judge’s weight); votes weighted by the participant’s accuracy on all items *except* the target item; and votes weighted by the participant’s cognitive reflection test (CRT) score.

To rigorously estimate what a live system could achieve, where it must learn a user’s reliability before applying it, we split the 40 items randomly into two halves of 10 true and 10 false items (1,500 stratified splits). We estimated each participant’s accuracy on the first half, and applied those weights and panel selections to their judgments on the second

half. Finally, we simulated crowds composed exclusively of Left-leaning participants, exclusively of Right-leaning participants, an exact half-and-half split, and random draws. All reported confidence intervals represent percentiles over simulations or bootstrap resamples of participants.

Effect of the community verdict. To measure social influence, we isolated every consecutive pair of items a participant judged. We flagged whether their answer to the first item placed them in the minority relative to that item’s final community vote (the verdict they had been shown, since the tally sent to participants was the final one; Section 3.4). We then modeled their answer on the *subsequent* item as a function of their own previous answer, their minority status, and the subsequent item’s actual veracity, clustered by participant.

Qualitative Analysis. We analyzed the interview transcripts and notes using reflexive thematic analysis [11, 12]. The first author coded all nine interviews across three passes: an initial pass structured by the interview guide, a second pass generating 30 open codes, and a final consolidation into four overarching themes accompanied by a complete re-reading of the data against the resulting codebook. We report the number of interviewees who made statements instantiating each code; codes were applied strictly to statements volunteered by the participant, not to ideas suggested by the interviewer that the participant merely agreed with. Following standard practices for single-coder reflexive analysis outlined by McDonald et al., we do not report an inter-rater reliability statistic [45]; instead, we include the full codebook with participant-level assignments in the supplementary material to allow for auditing. All quotes were translated from Spanish to English by the first author.

Positionality. The first author is Colombian, a native Spanish speaker, and a member of the community from which participants were recruited. He conducted the recruitment, the interviews, and the qualitative coding. While his standing in the community was the sole factor that made recruitment and trust-building possible, this insider status inherently shaped the data: participants knew him personally, several expressed a feeling of obligation to him, and interviewees may have framed their feedback more positively than they would have to an outside researcher. The second author is a computational social scientist who has extensively studied misinformation on WhatsApp across multiple countries, but is not a member of this specific community.

Data availability. The study stimuli, pseudonymized judgment data, participant demographics, interview notes, codebook, and analysis code are available in an anonymized repository.² Interview audio recordings are excluded, as participants did not consent to their release.

4 Quantitative Results

We first describe individual judgments (Section 4.1), followed by the specific content types the community failed to catch (Section 4.2), the role of partisan congruence and crowd composition (Section 4.3), the limits of aggregating and weighting expertise (Section 4.4), and how seeing the community verdict influenced subsequent behavior (Section 4.5). Appendix C (Table C.1) collects the estimates and tests behind each result, and Appendix D (Table D.1) reports the judgment-level models.

²https://drive.google.com/drive/folders/1tY4fykTsFqZaTPXuVt1Gv4GZj4PZpT6A?usp=drive_link

4.1 Individual Judgments: A Pervasive Default to Truth

Participants demonstrated a pervasive bias toward belief, proving highly capable of recognizing truth but significantly worse at detecting falsehoods. Overall, participants correctly evaluated 67% of the items they received. However, they successfully accepted 79% of the true items while managing to reject only 56% of the false ones. This vulnerability was consistent across the community: 29 of the 38 participants were significantly more accurate on true items than on false ones ($p < .001$; Appendix C, Table C.1). At the judgment level, the odds of a correct answer were nearly three times higher for true items than for false ones.

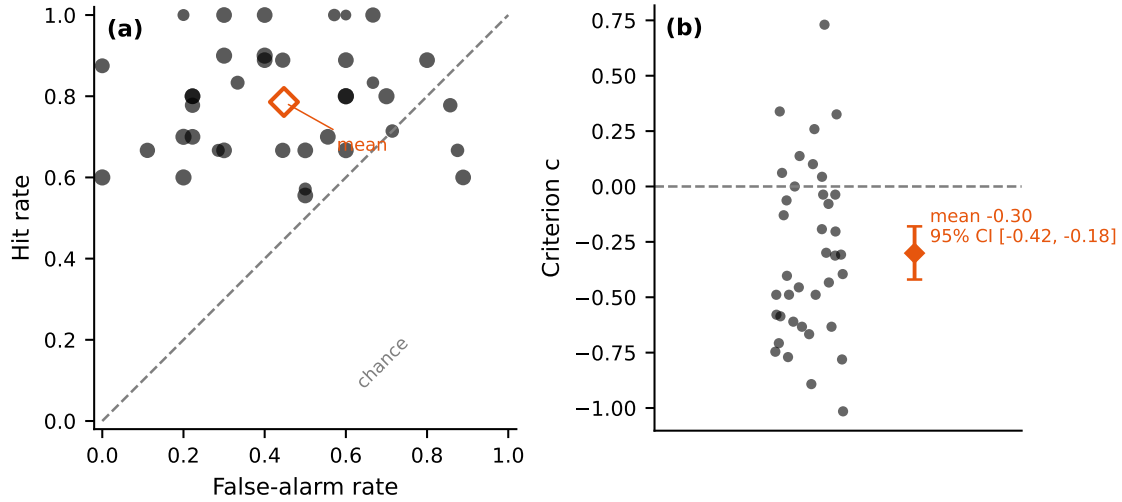


Fig. 3. Signal detection analysis of individual judgments. (a) Participant hit rates versus false-alarm rates, illustrating the pervasive bias toward accepting content as true. (b) Distribution of the response criterion (c), where negative values indicate a systemic default to trust.

Signal detection analysis (Figure 3) confirms that this dynamic is driven by a psychological default toward trust rather than an absolute inability to discern facts. Participants demonstrated modest sensitivity in distinguishing truth from fiction ($d' = 0.90$), but their judgments were heavily skewed by a response criterion shifted toward “true” ($c = -0.30$, $p < .001$). Because of this shifted criterion, 61% of all judgments cast by the community were “true,” despite half the stimulus set being false. Individual accuracy varied widely (ranging from 0.36 to 0.94), but this variation was not driven by standard demographics; a regression on age, education, political leaning, and cognitive reflection scores failed to significantly predict accuracy (Appendix D, Table D.2).

4.2 What the Community Missed: The Power of Video and Protective Warnings

Figure 4 illustrates how the community performed on specific items. Aggregation proved highly effective for factual content: the full community’s majority verdict was correct on 18 of the 20 true items. However, the majority caught only 12 of the 20 false items.

The seven false items that successfully deceived the majority clustered into two distinct categories: protective warning chains and political videos. Participants were fooled by half (4 of 8) of the false safety warnings, including

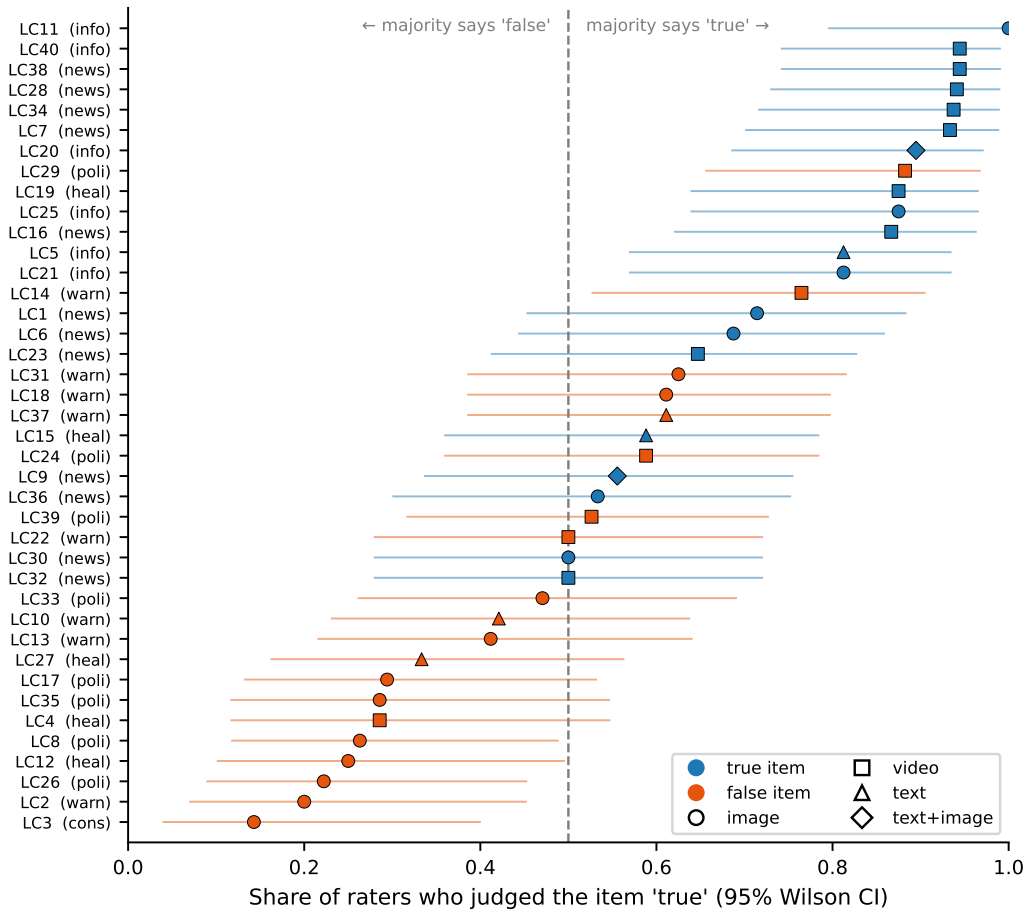


Fig. 4. Community accuracy across all 40 stimuli, ordered by the proportion of "true" judgments (with 95% Wilson intervals). True items (blue) were consistently accepted by the majority, while false items (orange) varied widely, with protective warning chains and political videos frequently deceiving the crowd.

a fictitious legal loophole to avoid paying road tolls (believed by 76%), a fabricated letter about robbers posing as government officials (62%), and a hoax about preventing ATM data theft (61%). They were also deceived by all three false political videos in the set, including a propaganda montage overlaying a foreign president's speech onto local politicians, which 88% of the community believed to be genuine. Conversely, every false political *image* (such as fabricated tweets and doctored charts) was successfully rejected by the majority.

Statistical modeling confirms this modality effect. Across the dataset, false videos were detected only 41% of the time, compared to a 63% detection rate for false images and text (item-level Welch $p = .049$). Judgment-level models confirm that the video format lowered the odds of catching a falsehood (Appendix D, Table D.1), suggesting that the realism of video overrides the skepticism participants applied to text and images. This effect rests on six false videos, two of which are the propaganda montages whose labels are debatable (Section 3.2); without those two it is smaller and not significant (Appendix F, Table F.1).

4.3 Partisan Congruence and the Vulnerability of Homogeneous Crowds

The data reveals that partisanship severely distorts community judgment. By isolating the six false items with a clear political slant and cross-referencing them with the 31 participants who explicitly identified with the Left or the Right, we found a massive congruence effect. When a false item flattered the rater's own political side, they believed it 51% of the time. When the falsehood favored the opposing side, they believed it only 19% of the time. Statistical modeling confirms that participants were 4.5 times more likely to believe a falsehood if it aligned with their political identity ($p = .009$). The congruent side believed the item more on each of the six items, and the effect holds under item-clustered models, without the three debatable items, and under the alternative slant coding (Appendix F, Table F.1).

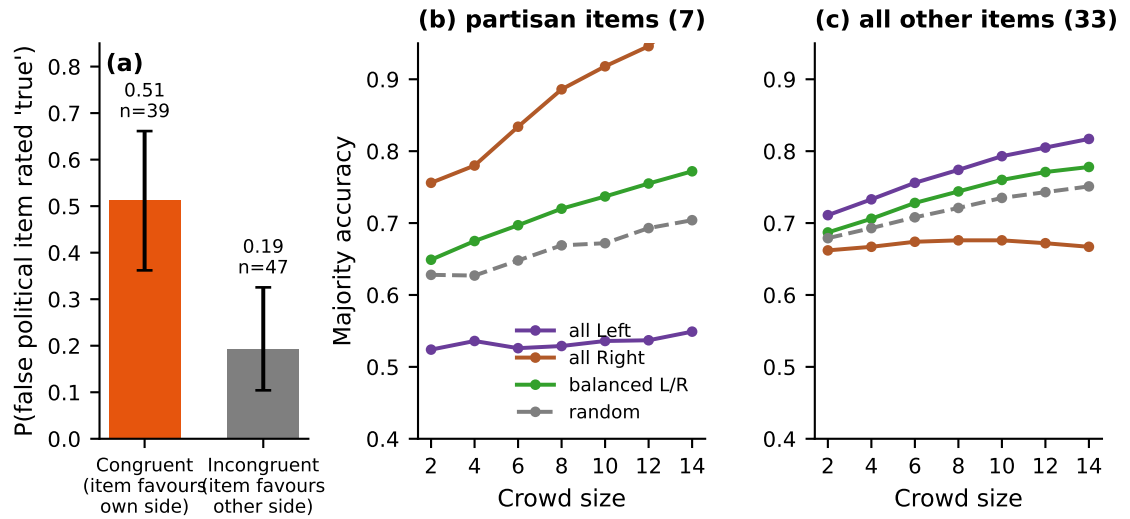


Fig. 5. The impact of partisan congruence and crowd composition on accuracy. (a) Participants were highly likely to believe false political items that favored their own side. (b, c) Simulated crowd accuracy on partisan and non-partisan items demonstrates that homogeneous groups amplify existing biases, while balanced groups stabilize accuracy.

This congruence effect has dire implications for decentralized moderation, as WhatsApp groups are often highly politically homogeneous. To test this, we simulated crowds drawn exclusively from our Left-identifying participants, exclusively from Right-identifying participants, and balanced half-and-half (Figure 5; Appendix E, Table E.1). On the partisan items, a homogeneous crowd's accuracy swung wildly depending on whether the falsehoods favored their side: an all-Left crowd of eight raters achieved only 53% accuracy, whereas an all-Right crowd achieved 89% on the same items (because the majority of the slanted falsehoods in our organic dataset happened to favor the Left). However, politically balanced crowds stabilized at 72% accuracy regardless of the item's slant. The lesson for system design is clear: a homogeneous crowd does not correct misinformation; it simply ratifies its own biases.

4.4 Aggregation and the Illusion of Identifying Expertise

If individual judgment is flawed, can aggregation save it? A simple, unweighted majority of the full community proved highly effective, correctly classifying 79% of the items (95% of the true items and 63% of the false ones; item-level 95%

interval 0.64 to 0.89). As shown in Figure 6a, most of the benefits of aggregation are realized quickly: majority accuracy reaches 71% with just seven raters and hits diminishing returns around 15 raters.

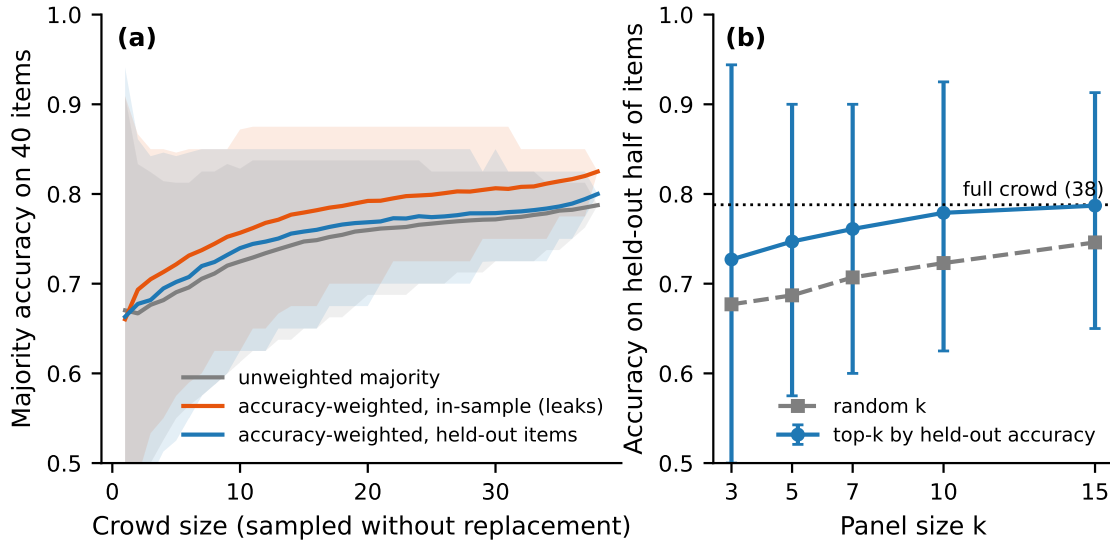


Fig. 6. Crowd aggregation and the limits of expertise weighting. (a) Simple majority voting reaches 79% accuracy with roughly 15 raters. (b) A panel of the raters most accurate on one half of the items improves only modestly on a random panel of the same size on the other half and does not exceed the full crowd, reflecting the instability of individual accuracy over a two-week window.

However, attempting to weight these votes by participant expertise proved futile in a short-term deployment. While a naive weighting model, scoring participants based on their accuracy across the entire study, appeared to boost crowd accuracy to 83%, this gain was an illusion. When we rigorously separated the data, estimating a participant’s “expertise” weight on one half of the items and applying it to the other half, the weighted majority gained less than one percentage point (+0.6%) over a simple unweighted vote.

This failure occurs because two weeks of binary judgments is simply not enough data to identify reliable experts. A participant’s accuracy in week one correlated with their accuracy in week two at just $r = 0.25$, and split-half reliability across random item splits averaged only $r = 0.23$. Consequently, any system that attempts to build reputation weights based on short-term participation will largely just amplify noise rather than expertise.

The contrast with items is instructive. Each item was judged by two disjoint groups of participants (the two schedules that contained it), and the share judging it true in one group correlated with the share in the other at $r = 0.69$ across the 40 items; the two groups’ majority verdicts agreed on 32 of the 40 (Appendix F, Table F.1). Sixteen to nineteen judgments were enough to measure how hard an item was, while seventeen judgments were not enough to measure how reliable a rater was. In this deployment, items replicated and raters did not, which argues for interventions aimed at content (Section 4.2) over interventions aimed at ranking raters.

4.5 The Community Verdict Lowered the Threshold for Belief, Not the Error Rate

We designed the bot to show participants the community’s aggregate verdict after they cast their vote, hypothesizing that this social feedback might help less accurate users calibrate over time. The behavioral data show a different effect:

being contradicted made participants more likely to accept the next item as true, whichever way the community had voted.

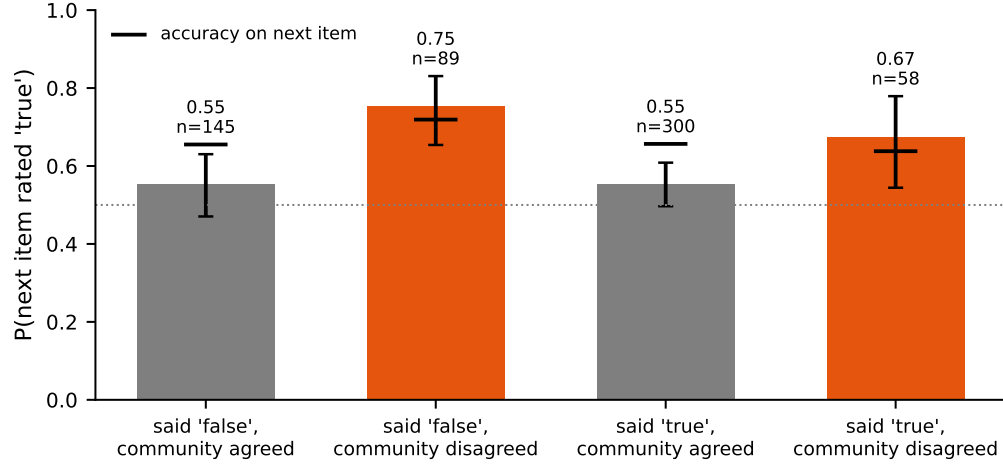


Fig. 7. The behavioral effect of social feedback. Disagreeing with the community majority increased a participant's likelihood of answering "true" on the next item whether their own answer had been "false" or "true", so the shift was toward acceptance rather than toward the community's position, and it did not improve accuracy (black ticks).

In 25% of the item pairs, participants discovered that their first answer placed them in the minority. Controlling for their own previous answer and for the veracity of the next item, having been in the minority raised the odds of answering "true" on the next item by a factor of 1.74 ($p = .001$; 1.65 when clustering on items, Appendix D, Table D.1). The shift was not toward the community's position. It appeared among participants who had answered "false" and learned that the community said "true" (75% answered "true" next, against 55% after an agreement) and equally among those who had answered "true" and learned that the community said "false" (67% against 55%). The probability that the next answer matched the next item's eventual majority was the same after a disagreement as after an agreement (74% in both cases, $p = .47$), and when the next item's majority was "false", participants who had just been contradicted answered "true" more often than those who had not (45% versus 26%), moving away from the majority. Being contradicted by peers lowered the threshold for accepting content, the same direction as the truth bias in Section 4.1 and the congruence effect in Section 4.3.

This shift did not improve fact-checking. The probability of answering the next item correctly was unaffected by whether the participant had just agreed or disagreed with the majority ($p = .60$), and the shift appeared at every accuracy level, including the top tercile. Showing the community verdict did not teach participants to spot falsehoods; being told that they had disagreed made them more accepting.

5 Qualitative Findings

We analyzed the nine interviews across 30 codes, consolidating them into four overarching themes (Appendix G, Table G.1). The full qualitative dataset, including participant-level assignments and original Spanish transcripts, is also available in the supplementary material. Below, we report the number of interviewees whose accounts instantiated each code to provide a transparent measure of prevalence.

5.1 Heuristics, Friction, and Social Consultation

When deciding what was true, participants rarely relied on rigorous, independent verification. Instead, they navigated judgments using a blend of lived experience, political identity, and social consultation. Six of the nine interviewees evaluated items against their own prior knowledge or local awareness. This localized heuristic was often highly accurate: P07, who lives in a small town, correctly dismissed every kidnapping alert because real disappearances leave unavoidable social evidence (*"You see the families worried... how can children be being stolen and nobody even knows which child was stolen?"*).

When lived experience fell short, participants frequently outsourced the cognitive labor to their immediate social circles (five interviewees). P01 asked whoever was nearby, P06 debated items with classmates, and P09 consulted a colleague. However, this social consultation did not guarantee accuracy; P03 read a scam-warning chain to her husband because he *"is more informed,"* and submitted his incorrect verdict of *"true."*

Only six interviewees reported leaving the app to search Google or news sites, and two of them did so only once. For several participants, the user interface and task friction actively discouraged external verification. P01 feared she would forget to finish the task if she closed WhatsApp to open a browser, while P03 admitted she answered *"true"* to a video that failed to load simply so she wouldn't risk losing her financial incentive. Others bypassed verification entirely by relying on surface cues, P05 assumed well-produced content was real because it *"looks more like an advertisement,"* or applied rigid political filters. P04 classified anything that read as propaganda from either side as false, while P03 rejected a claim about previous governments being worse simply because, in her view, *"now we are worse off."*

5.2 The Dual Meaning of Social Feedback

We hypothesized that showing participants the community's verdict would help them calibrate their accuracy. Instead, the feedback acted as a psychological mirror that participants interpreted in highly divergent ways. For six interviewees, agreeing with the majority provided comforting confirmation, giving P09 *"a feeling that it was the true one, that a consensus was being built."*

Disagreement, however, fractured the group. Five participants internalized it as a cue for self-doubt: P02 *"would read it again to see what had happened,"* P08 wondered *"is it right, or am I wrong?,"* and P05 concluded he *"wasn't well informed."* For others, the exchange felt gamified, with P04 and P06 treating disagreement like a loss (*"Damn, I lost that one,"* P06).

Conversely, four participants externalized the disagreement, using it to judge the community rather than themselves. P01 assumed the rest of the group lacked knowledge; P03 inferred that the community was skewed toward young supporters of the opposing political party; and P09 decided that the community *"does not represent me."* Three of these participants explicitly claimed they ignored the community verdict entirely (*"I knew what I had put and I believed in what I had put, and that was it,"* P07). However, the quantitative behavioral data (Section 4.5) directly contradicts these self-reports: participants at every accuracy level, including the highest performers who claimed to be unmoved, were more likely to answer *"true"* on the item that followed a disagreement.

Finally, trust in the aggregate was highly conditional. P09 noted he would only trust a community verdict if he could be guaranteed the group was *"very heterogeneous... in age, schooling, professional level."* The feedback UI itself also caused minor confusion; P08 was unsure if *"80% true"* meant her answer was correct or if it simply meant most people voted that way, and wondered if *"the community"* referred to other study participants or the original group chat the content came from.

5.3 Motivations and Design Requirements for Deployment

When asked why they participated and what a real, non-compensated version of the system would require, participants highlighted strict boundaries for trust and engagement.

While six interviewees cited the financial payment as a primary motivation, the local relevance of the task was equally vital (five interviewees). P01 was pleasantly surprised that a study run by a foreign university surfaced news about her own town, and P03 valued that “*everything was based on what is really happening*.” Because participants could fit the WhatsApp-based task seamlessly into their daily routines (nine interviewees), the friction was low enough to sustain engagement.

However, participants warned that launching an uncompensated system faces a massive trust barrier. P01 only joined the study because a trusted relative vouched for the researcher, and P07 noted that her relative ignored the recruitment message entirely “*because she was afraid it was a scam*.” They suggested a live system would require backing from a highly trusted entity, such as the State or WhatsApp itself, and would need to be framed around a tangible community good, such as scam prevention.

Furthermore, participants demanded that their labor result in visible outcomes. P03 stated she would not contribute “*words the wind carries away*,” but would participate daily if the votes actually led to visible warnings or interventions. Regarding cadence, six interviewees stressed that the system must respect user bandwidth, recommending a maximum of one or two items a day with a clear 24-hour response window (“*then you can plan*,” P09). Ultimately, while participants found the WhatsApp interface exceptionally easy to use, preferring a simple A/B reply over external web forms they recognized that the platform currently lacks the visual infrastructure, such as the community warning overlays seen on Facebook, required to make their peer moderation visible.

6 Discussion

The promise of crowdsourced moderation relies on the assumption that laypeople, when aggregated, can reliably discern truth from fiction. Our field deployment demonstrates that while this holds for basic factual claims, the unique architecture and relational dynamics of encrypted messaging violently disrupt this mechanism. Below, we discuss how network homophily, the psychology of trust, and the pitfalls of social feedback require a fundamental rethinking of how we design community moderation for private spaces.

6.1 Homophily and Cross-Group Aggregation

Our results show that the accuracy of a localized crowd on partisan content depends heavily on its partisan composition. Because WhatsApp groups are anchored in real-world, homophilous networks (e.g., family chats, church groups), they lack the ideological diversity required to neutralize partisan bias. As our simulations demonstrated, a homogeneous crowd does not fact-check its own side’s propaganda; it simply ratifies it. Consequently, state-of-the-art moderation algorithms that rely on “bridging” across ideological divides, such as X’s Community Notes [68] will inherently fail if restricted to a single WhatsApp group, as there is no out-group available to bridge with.

Consequently, effective moderation in encrypted spaces cannot rely solely on intra-group voting; platforms must facilitate *cross-group aggregation*. By relying on privacy-preserving hashing (where the platform aggregates votes on a viral media hash without reading the content itself), designers can pool judgments across thousands of disparate, encrypted communities. This global pooling generates the ideological diversity necessary for bridging algorithms to function. Furthermore, because a single group’s consensus is likely biased, the UI should shift away from declaring

a definitive “True/False” verdict. Instead, interfaces should highlight *contestedness*, for instance, warning a user that “While your group voted this true, it is highly contested by other communities.”

6.2 Modality Effects and Targeted Friction

We observed a pervasive default to truth across the community, but this vulnerability was not uniformly distributed. The crowd struggled significantly on two specific vectors: visual media (false videos) and relational warnings (fake kidnapping or scam alerts). In interviews, it became clear that users circulate these warnings not out of malice, but out of a localized duty of care to protect their families.

Because users apply different levels of skepticism to different types of content, platforms must intervene at the modality level. Rather than applying generic friction to all forwarded messages, WhatsApp could introduce targeted friction, such as mandatory preview screens or forced delays, specifically for forwarded videos and text-based safety chains. Furthermore, designers can leverage this protective instinct rather than fighting it. Currently, most automated fact-checks are framed as a “correction,” which triggers defensive, partisan reactions. If a system instead frames peer feedback as a protective shield, e.g., “Other communities flagged this video as a scam to protect their groups”, it aligns the UI with the users’ own motivations, making them far more receptive to the intervention.

6.3 Reputation Weighting and Social Feedback

When designing crowdsourced systems, there is a strong temptation to gamify the process: weighting the votes of “expert” users and showing live community feedback to help users calibrate. Our deployment proves that both of these mechanisms backfire in the context of encrypted chat.

First, attempting to weight votes based on short-term accuracy is a mathematical dead end. Over our two-week deployment, individual accuracy was too noisy ($r = 0.23$) to reliably identify experts, meaning any system that quickly assigns reputation scores will merely amplify noise. Designers must delay reputation weighting, gathering reliability data over months rather than weeks before allowing a user’s weight to influence the aggregate score.

Second, exposing users to the live community verdict did not calibrate them. Discovering they were in the minority increased a participant’s likelihood of answering “true” on the next item by a factor of 1.7, whichever way the community had voted, without improving their accuracy. In a community that already leans toward belief, this is the worst direction for feedback to push: the verdict most often contradicted the skeptics, and the contradiction lowered their skepticism on the next item.

Therefore, a peer moderation system on WhatsApp should not show bare verdicts between judgments. If a verdict shown after one item shifts the answer to the next, and those answers then feed the labels or reputation scores the system computes, the aggregate drifts toward acceptance rather than toward independent judgment. Verdicts should be delayed until a participant’s judgments for the day are in, or shown together with the reasons raters gave, which give a contradicted participant something to evaluate rather than a pull to accept.

7 Limitations

Our findings must be interpreted within the constraints of our methodology. First, while our two-week deployment allowed for deep, mixed-methods analysis of a specific community, the short timeframe and sample size limit our ability to model long-term behavioral changes. Furthermore, as noted in our methodology, we fixed the order of items within each schedule to ensure consistent social feedback dynamics across the cohort. Because position and item were confounded, we could not isolate longitudinal fatigue or learning effects. Three of the twenty false items are misleading

rather than fabricated, and the partisan slant of items was coded by the authors; Appendix F (Table F.1) shows which results depend on these choices.

Second, although we achieved a high degree of ecological validity by using organic viral media donated directly from participants' own groups, the act of evaluating content via a bot for financial compensation remains artificial. In reality, users encounter misinformation passively while scrolling through rapid, multi-threaded family or community chats. While the financial incentive was necessary to sustain daily engagement for the study, it likely motivated participants to complete evaluations they might otherwise have ignored, altering their natural friction thresholds.

Finally, our field deployment was situated in Colombia, a context characterized by specific relational hierarchies, deep political polarization, and a high reliance on WhatsApp for daily civic infrastructure. While the cognitive heuristics we observed, such as the truth bias and the modality effect of video are deeply rooted in human psychology, the specific social dynamics of peer correction and conformity are inherently culturally mediated. Future work must test these moderation architectures across diverse cultural contexts to understand how local relational norms shape community fact-checking.

8 Conclusion

This study provides the first in-situ empirical evaluation of crowdsourced fact-checking within an encrypted messaging environment. By deploying a custom moderation bot directly inside a Colombian WhatsApp community, we demonstrated that the mechanisms driving crowd wisdom on public platforms fundamentally fracture within private, relational networks.

Participants exhibited a pervasive default to truth, struggling significantly to reject falsehoods when encountering visual media and protective safety warnings. We found that in politically homogeneous spaces, decentralized crowds do not correct misinformation; they merely amplify their existing biases. Displaying the live peer verdict did not calibrate participants: being contradicted made them more accepting of the next item without making them more accurate.

If platforms are to protect communities without compromising end-to-end encryption, designers must abandon the assumptions built for public social media. Future moderation architectures for private chat must facilitate cross-group aggregation to neutralize network homophily, withhold bare peer verdicts between judgments, and apply targeted friction to the specific modalities that bypass human skepticism.

Acknowledgements and Generative AI usage

We thank the study participants for their time and candor. Generative AI tools like ChatGPT were used to edit the prose of this manuscript and in some of the data analysis. They were not used for study design, data collection, coding of interviews, or interpretation. All claims and citations in the paper were verified by the authors.

References

- [1] Dhruv Agarwal, Farhana Shahid, and Aditya Vashistha. 2024. Conversational agents to facilitate deliberation on harmful content in WhatsApp groups. *Proceedings of the ACM on human-computer interaction* 8, CSCW2 (2024), 1–32.
- [2] Jennifer Allen, Antonio A Arechar, Gordon Pennycook, and David G Rand. 2021. Scaling up fact-checking using the wisdom of crowds. *Science advances* 7, 36 (2021), eabf4393.
- [3] Jennifer Allen, Cameron Martel, and David G. Rand. 2022. Birds of a feather don't fact-check each other: Partisanship and the evaluation of news in Twitter's Birdwatch crowdsourced fact-checking program. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 245, 19 pages. doi:10.1145/3491102.3502040
- [4] Chinmayi Arun. 2019. On WhatsApp, Rumours, and Lynchings. *Economic & Political Weekly* 54, 6 (2019).

Manuscript submitted to ACM

- [5] R. Harald Baayen, Douglas J. Davidson, and Douglas M. Bates. 2008. Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language* 59, 4 (2008), 390–412. doi:10.1016/j.jml.2007.12.005
- [6] Shakuntala Banaji, Ramnath Bhat, Anushi Agarwal, Nihal Passanha, and Mukti Sadhana Pravin. 2019. *WhatsApp vigilantes: An exploration of citizen reception and circulation of WhatsApp misinformation linked to mob violence in India*. Report. Department of Media and Communications, London School of Economics and Political Science.
- [7] Cédric Batailler, Skylar M. Brannon, Paul E. Teas, and Bertram Gawronski. 2022. A Signal Detection Approach to Understanding the Identification of Fake News. *Perspectives on Psychological Science* 17, 1 (2022), 78–98. doi:10.1177/1745691620986135
- [8] Joshua Becker, Ethan Porter, and Damon Centola. 2019. The wisdom of partisan crowds. *Proceedings of the National Academy of Sciences* 116, 22 (2019), 10717–10722.
- [9] Md Momen Bhuiyan, Amy X. Zhang, Connie Moon Sehat, and Tanushree Mitra. 2020. Investigating Differences in Crowdsourced News Credibility Assessment: Raters, Tasks, and Expert Criteria. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW2, Article 93 (Oct. 2020), 26 pages. doi:10.1145/3415164
- [10] Nadia M. Brashier and Elizabeth J. Marsh. 2020. Judging Truth. *Annual Review of Psychology* 71 (2020), 499–515. doi:10.1146/annurev-psych-010419-050807
- [11] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101. doi:10.1191/1478088706qp063oa
- [12] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health* 11, 4 (2019), 589–597. doi:10.1080/2159676X.2019.1628806
- [13] Yuwei Chuai, Haoye Tian, Nicolas Pröllochs, and Gabriele Lenzini. 2024. Did the roll-out of community notes reduce engagement with misinformation on X/Twitter? *Proceedings of the ACM on human-computer interaction* 8, CSCW2 (2024), 1–52.
- [14] Tim Draws, David La Barbera, Michael Soprano, Kevin Roitero, Davide Ceolin, Alessandro Checco, and Stefano Mizzaro. 2022. The effects of crowd worker biases in fact-checking tasks. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*. Association for Computing Machinery, New York, NY, USA, 2114–2124.
- [15] Chiara Patricia Drolsbach and Nicolas Pröllochs. 2023. Diffusion of Community Fact-Checked Misinformation on Twitter. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 267 (Oct. 2023), 22 pages. doi:10.1145/3610058
- [16] Ullrich K. H. Ecker, Stephan Lewandowsky, John Cook, Philipp Schmid, Lisa K. Fazio, Nadia Brashier, Panayioti Kendeou, Emily K. Vraga, and Michelle A. Amazeen. 2022. The psychological drivers of misinformation belief and its resistance to correction. *Nature Reviews Psychology* 1, 1 (2022), 13–29. doi:10.1038/s44159-021-00006-y
- [17] Ziv Epstein, Gordon Pennycook, and David Rand. 2020. Will the Crowd Game the Algorithm? Using Layperson Judgments to Combat Misinformation on Social Media by Downranking Distrusted Sources. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3313831.3376232
- [18] Martin Flinham, Christian Karner, Khaled Bachour, Helen Creswick, Neha Gupta, and Stuart Moran. 2018. Falling for fake news: investigating the consumption of news via social media. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–10.
- [19] Kiran Garimella. 2025. Community-Driven Fact-Checking on WhatsApp: Who Fact-Checks Whom, Why, and With What Effect? *Computational Approaches to Content Moderation and Platform Governance workshop at ICWSM* (2025).
- [20] Kiran Garimella and Simon Chauchard. 2024. WhatsApp explorer: A data donation tool to facilitate research on WhatsApp. *Mobile Media & Communication* (2024), 20501579251326809.
- [21] Kiran Garimella, Princessa Cintaquia, Juan José Rojas-Constain, Bharat Kumar Nayak, and Aditya Vashistha. 2025. Global Patterns of Viral Content on WhatsApp. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 19. 586–601.
- [22] Kiran Garimella and Dean Eckles. 2020. Images and misinformation in political groups: Evidence from WhatsApp in India. *Harvard Kennedy School Misinformation Review* (2020).
- [23] Bertram Gawronski, Nyx L. Ng, and Dillon M. Luke. 2023. Truth sensitivity and partisan bias in responses to misinformation. *Journal of Experimental Psychology: General* 152, 8 (2023), 2205–2236. doi:10.1037/xge0001381
- [24] Christine Geeng, Savanna Yee, and Franziska Roesner. 2020. Fake news on Facebook and Twitter: Investigating how people (don't) investigate. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–14.
- [25] William Godel, Zeve Sanderson, Kevin Aslett, Jonathan Nagler, Richard Bonneau, Nathaniel Persily, and Joshua A Tucker. 2021. Moderating with the mob: Evaluating the efficacy of real-time crowdsourced fact-checking. *Journal of Online Trust and Safety* 1, 1 (2021).
- [26] Bernard Grofman, Guillermo Owen, and Scott L. Feld. 1983. Thirteen theorems in search of the truth. *Theory and Decision* 15, 3 (1983), 261–278. doi:10.1007/BF00125672
- [27] Douglas Guilbeault, Joshua Becker, and Damon Centola. 2018. Social learning and partisan bias in the interpretation of climate trends. *Proceedings of the National Academy of Sciences* 115, 39 (2018), 9714–9719.
- [28] Jacob Gursky, Martin J Riedl, Katie Joseff, and Samuel Woolley. 2022. Chat apps and cascade logic: A multi-platform perspective on India, Mexico, and the United States. *Social Media+ Society* 8, 2 (2022), 20563051221094773.
- [29] Michael Hameleers, Thomas E. Powell, Toni G. L. A. Van Der Meer, and Lieke Bos. 2020. A Picture Paints a Thousand Lies? The Effects and Mechanisms of Multimodal Disinformation and Rebuttals Disseminated via Social Media. *Political Communication* 37, 2 (2020), 281–301. doi:10.1080/10584609.2019.1674979

- [30] Michael J. Hautus. 1995. Corrections for extreme proportions and their biasing effects on estimated values of d' . *Behavior Research Methods, Instruments, & Computers* 27, 1 (1995), 46–51. doi:10.3758/BF03203619
- [31] Farnaz Jahanbakhsh, Amy X Zhang, Adam J Berinsky, Gordon Pennycook, David G Rand, and David R Karger. 2021. Exploring lightweight interventions at posting time to reduce the sharing of misinformation on social media. *Proceedings of the ACM on human-computer interaction* 5, CSCW1 (2021), 1–42.
- [32] Bertrand Jayles, Hye-rin Kim, Ramón Escobedo, Stéphane Cezera, Adrien Blanchet, Tatsuya Kameda, Clément Sire, and Guy Theraulaz. 2017. How social information can improve estimation accuracy in human groups. *Proceedings of the National Academy of Sciences* 114, 47 (2017), 12620–12625. doi:10.1073/pnas.1703695114
- [33] Charles M. Judd, Jacob Westfall, and David A. Kenny. 2012. Treating stimuli as a random factor in social psychology: A new and comprehensive solution to a pervasive but largely ignored problem. *Journal of Personality and Social Psychology* 103, 1 (2012), 54–69. doi:10.1037/a0028347
- [34] Ashkan Kazemi, Kiran Garimella, Gautam Kishore Shahi, Devin Gaffney, and Scott A Hale. 2022. Research note: Tiplines to uncover misinformation on encrypted platforms: A case study of the 2019 Indian general election on WhatsApp. *Harvard Kennedy School Misinformation Review* 3, 1 (2022).
- [35] Prerana Khatiwada, Luke Halko, Nabiha Syed, Ashrey Mahesh, Anesh Alvanpour, and Matthew Louis Mauriello. 2025. Spotting Online News: A Mixed Method Study of Online News Engagement and Perceptions on Misinformation Interventions. *Proceedings of the ACM on Human-Computer Interaction* 9, CSCW1 (2025), 1–30.
- [36] Jooyeon Kim, Behzad Tabibian, Alice Oh, Bernhard Schölkopf, and Manuel Gomez-Rodriguez. 2018. Leveraging the Crowd to Detect and Reduce the Spread of Fake News and Misinformation. In *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining* (Marina Del Rey, CA, USA) (WSDM '18). Association for Computing Machinery, New York, NY, USA, 324–332. doi:10.1145/3159652.3159734
- [37] Neta Kliger-Vilenchik. 2022. Collective social correction: addressing misinformation through group practices of information verification on WhatsApp. *Digital Journalism* 10, 2 (2022), 300–318.
- [38] Timothy R. Levine. 2014. Truth-Default Theory (TDT): A Theory of Human Deception and Deception Detection. *Journal of Language and Social Psychology* 33, 4 (2014), 378–392. doi:10.1177/0261927X14535916
- [39] Jan Lorenz, Heiko Rauhut, Frank Schweitzer, and Dirk Helbing. 2011. How social influence can undermine the wisdom of crowd effect. *Proceedings of the National Academy of Sciences* 108, 22 (2011), 9020–9025. doi:10.1073/pnas.1008636108
- [40] Neil A. Macmillan and C. Douglas Creelman. 2005. *Detection Theory: A User's Guide* (2nd ed.). Lawrence Erlbaum Associates, Mahwah, NJ, USA. doi:10.4324/9781410611147
- [41] Pranav Malhotra. 2020. A Relationship-Centered and Culturally Informed Approach to Studying Misinformation on COVID-19. *Social Media + Society* 6, 3 (2020). doi:10.1177/2056305120948224
- [42] Pranav Malhotra. 2024. Misinformation in WhatsApp family groups: Generational perceptions and correction considerations in a meso-news space. *Digital journalism* 12, 5 (2024), 594–612.
- [43] Drew B. Margolin, Aniko Hannak, and Ingmar Weber. 2018. Political Fact-Checking on Twitter: When Do Corrections Have an Effect? *Political Communication* 35, 2 (2018), 196–219. doi:10.1080/10584609.2017.1334018
- [44] Cameron Martel, Jennifer Allen, Gordon Pennycook, and David G Rand. 2024. Crowds can effectively identify misinformation at scale. *Perspectives on Psychological Science* 19, 2 (2024), 477–488.
- [45] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and Inter-rater Reliability in Qualitative Research: Norms and Guidelines for CSCW and HCI Practice. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW, Article 72 (Nov. 2019), 23 pages. doi:10.1145/3359174
- [46] Philipe Melo, Johnnatan Messias, Gustavo Resende, Kiran Garimella, Jussara Almeida, and Fabricio Benevenuto. 2019. WhatsApp Monitor: A Fact-Checking System for WhatsApp. In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2019, Vol. 13)*. AAAI Press, 676–677. doi:10.1609/icwsm.v13i01.3271
- [47] Miriam J. Metzger, Andrew J. Flanagin, and Ryan B. Medders. 2010. Social and Heuristic Approaches to Credibility Evaluation Online. *Journal of Communication* 60, 3 (2010), 413–439. doi:10.1111/j.1460-2466.2010.01488.x
- [48] Nic Newman, A Ross Arguedas, Craig T Robertson, Rasmus Kleis Nielsen, and Richard Fletcher. 2025. *Digital news report 2025*. Reuters Institute for the study of Journalism.
- [49] Shmuel Nitzan and Jacob Paroush. 1982. Optimal Decision Rules in Uncertain Dichotomous Choice Situations. *International Economic Review* 23, 2 (1982), 289–297. doi:10.2307/2526438
- [50] Irene V Pasquetto, Eaman Jahani, Shubham Atreja, and Matthew Baum. 2022. Social debunking of misinformation on WhatsApp: the case for strong and in-group ties. *Proceedings of the ACM on human-computer interaction* 6, CSCW1 (2022), 1–35.
- [51] Gordon Pennycook and David G. Rand. 2019. Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. *Cognition* 188 (2019), 39–50. doi:10.1016/j.cognition.2018.06.011
- [52] Gordon Pennycook and David G. Rand. 2021. The Psychology of Fake News. *Trends in Cognitive Sciences* 25, 5 (2021), 388–402. doi:10.1016/j.tics.2021.02.007
- [53] Nicolas Pröllochs. 2022. Community-Based Fact-Checking on Twitter's Birdwatch Platform. In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2022, Vol. 16)*. AAAI Press, 794–805. doi:10.1609/icwsm.v16i1.19335
- [54] Julio C. S. Reis, Philipe Melo, Kiran Garimella, and Fabricio Benevenuto. 2020. Can WhatsApp benefit from debunked fact-checked stories to reduce misinformation? *Harvard Kennedy School Misinformation Review* 1, 5 (Aug. 2020). doi:10.37016/mr-2020-035

- [55] Gustavo Resende, Philipe Melo, Hugo Sousa, Johnnatan Messias, Marisa Vasconcelos, Jussara Almeida, and Fabrício Benevenuto. 2019. (Mis)Information Dissemination in WhatsApp: Gathering, Analyzing and Countermeasures. In *The World Wide Web Conference* (San Francisco, CA, USA) (*WWW '19*). Association for Computing Machinery, New York, NY, USA, 818–828. doi:10.1145/3308558.3313688
- [56] Patricia Rossini, Jennifer Stromer-Galley, Erica Anita Baptista, and Vanessa Veiga de Oliveira. 2021. Dysfunctional information sharing on WhatsApp and Facebook: The role of political talk, cross-cutting exposure and social corrections. *New media & society* 23, 8 (2021), 2430–2451.
- [57] Mohammed Saeed, Nicolas Traub, Maelle Nicolas, Gianluca Demartini, and Paolo Papotti. 2022. Crowdsourced Fact-Checking at Twitter: How Does the Crowd Compare With Experts?. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management* (Atlanta, GA, USA) (*CIKM '22*). Association for Computing Machinery, New York, NY, USA, 1736–1746. doi:10.1145/3511808.3557279
- [58] Emily Saltz, Claire R Leibowicz, and Claire Wardle. 2021. Encounters with visual misinformation and labels across platforms: An interview and diary study to inform ecosystem approaches to misinformation interventions. In *Extended abstracts of the 2021 CHI conference on human factors in computing systems*. 1–6.
- [59] Farhana Shahid, Dhruv Agarwal, and Aditya Vashistha. 2025. One Style Does Not Regulate All: Moderation Practices in Public and Private WhatsApp Groups. *Proceedings of the ACM on Human-Computer Interaction* 9, CSCW1 (2025), 1–30.
- [60] Isaac Slaughter, Axel Peytavin, Johan Ugander, and Martin Saveski. 2025. Community notes reduce engagement with and diffusion of false information online. *Proceedings of the National Academy of Sciences* 122, 38 (2025), e2503413122.
- [61] S. Shyam Sundar, Maria D. Molina, and Eugene Cho. 2021. Seeing Is Believing: Is Video Modality More Powerful in Spreading Fake News via Online Messaging Apps? *Journal of Computer-Mediated Communication* 26, 6 (2021), 301–319. doi:10.1093/jcmc/zmab010
- [62] James Surowiecki. 2004. *The Wisdom of Crowds: Why the Many Are Smarter Than the Few and How Collective Wisdom Shapes Business, Economies, Societies and Nations*. Doubleday, New York, NY, USA.
- [63] Inga K Trauthig, Katlyn Glover, Zelly Martin, Anastasia Goodwin, and Samuel Woolley. 2022. *Polarized Information Ecosystems and Encrypted Messaging Apps: Insights into Egypt, Ethiopia, and Libya*. Technical Report. Center for Media Engagement, University of Texas at Austin.
- [64] Rama Adithya Varanasi, Joyojeet Pal, and Aditya Vashistha. 2022. Accost, accede, or amplify: attitudes towards COVID-19 misinformation on WhatsApp in India. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [65] Santosh Vijaykumar, Yan Jin, Daniel Rogerson, Xuerong Lu, Swati Sharma, Anna Maughan, Bianca Fadel, Mariella Silva de Oliveira Costa, Claudia Pagliari, and Daniel Morris. 2021. How shades of truth and age affect responses to COVID-19 (Mis)information: randomized survey experiment among WhatsApp users in UK and Brazil. *Humanities and Social Sciences Communications* 8, 1, Article 88 (2021). doi:10.1057/s41599-021-00752-7
- [66] Joyce Vissenberg, Diego Martinez, Natalia Edisherashvili, Marit Puusepp, Łukasz Tomczyk, Veronica Donoso, and Leen d’Haenens. 2025. *Advancing media literacy & digital skills interventions: recommendations for stakeholders*. Project deliverable D4.1. KU Leuven, REMEDIS project. doi:10.5281/zenodo.15879417
- [67] Chloe Wittenberg, Ben M. Tappin, Adam J. Berinsky, and David G. Rand. 2021. The (minimal) persuasive advantage of political video over text. *Proceedings of the National Academy of Sciences* 118, 47 (2021), e2114388118. doi:10.1073/pnas.2114388118
- [68] Stefan Wojcik, Sophie Hilgard, Nick Judd, Delia Mocanu, Stephen Ragain, M. B. Fallin Hunzaker, Keith Coleman, and Jay Baxter. 2022. Birdwatch: Crowd Wisdom and Bridging Algorithms can Inform Understanding and Reduce the Spread of Misinformation. arXiv:2210.15723

A Participants

Table A.1 lists the 38 participants with the covariates from the pre-survey and their engagement in the study. Political leaning is the group of national politicians the participant said they identified with. CRT is the number of correct answers (0 to 3) on a three-item cognitive reflection test. “Items” is the number of the 20 assigned items the participant answered, and “Accuracy” the share of those answers that matched the ground truth. Two participants (P15, P23) answered items but did not complete the pre-survey; they are included in all accuracy analyses and excluded from analyses that use covariates. Participants P01 to P09 are the nine interviewees (Section 3.5).

Table A.1. Participants. Gender, age, leaning, education, time since last formal study, internet use (1 = rarely, 5 = very often), religiosity, and CRT score are from the pre-survey; Items and Accuracy are from the judgment data.

ID	Gender	Age	Leaning	Educ. (yrs)	Last study	Internet	Religious	CRT	Items	Accuracy
P01	Female	52	Right	10	No info	5	Yes	0	20	0.80
P02	Female	21	Left	9	1–5 years	4	Yes	3	19	0.68
P03	Female	47	Right	11	> 10 years	5	Yes	3	19	0.53
P04	Female	21	Right	18	< 1 year	5	No	3	13	0.69
P05	Male	26	Left	19	< 1 year	4	No	3	19	0.53
P06	Male	21	Right	12	< 1 year	5	Not answered	1	14	0.36

ID	Gender	Age	Leaning	Educ. (yrs)	Last study	Internet	Religious	CRT	Items	Accuracy
P07	Female	27	Center	17	< 1 year	5	Yes	0	20	0.70
P08	Female	22	Left	15	< 1 year	4	Yes	3	18	0.78
P09	Male	27	Left	21	< 1 year	2	No	2	15	0.73
P10	Female	36	Left	11	> 10 years	5	Yes	0	19	0.37
P11	Female	64	Right	11	> 10 years	5	Yes	1	16	0.50
P12	Female	24	Center	17	< 1 year	5	No	3	14	0.50
P13	Male	19	Center	12	< 1 year	4	Yes	3	19	0.53
P14	Female	59	Left	14	> 10 years	3	Not answered	3	13	0.54
P15	Pre-survey not completed								12	0.58
P16	Male	57	Left	15	> 10 years	3	Yes	3	20	0.55
P17	Male	18	Left	13	< 1 year	5	Yes	0	19	0.58
P18	Male	53	Right	11	5–10 years	5	Yes	1	19	0.58
P19	Male	19	Center	12	< 1 year	5	No	2	20	0.60
P20	Female	21	Right	2	1–5 years	5	Yes	1	20	0.60
P21	Male	22	Left	13	< 1 year	4	Yes	1	18	0.61
P22	Female	36	Right	5	> 10 years	5	No	3	19	0.63
P23	Pre-survey not completed								9	0.67
P24	Male	62	Left	15	> 10 years	5	No	2	12	0.67
P25	Male	22	Center	14	< 1 year	1	Yes	1	18	0.67
P26	Male	36	Right	11	> 10 years	4	Yes	1	18	0.72
P27	Male	20	Left	16	< 1 year	3	Yes	2	19	0.74
P28	Female	46	Right	11	1–5 years	5	Yes	2	19	0.74
P29	Male	20	Left	2	< 1 year	3	Yes	3	20	0.75
P30	Female	52	Right	20	> 10 years	5	Yes	0	20	0.75
P31	Female	45	Right	13	< 1 year	5	Yes	0	18	0.78
P32	Female	18	Left	13	< 1 year	5	Yes	2	19	0.79
P33	Female	18	Left	13	< 1 year	5	Yes	3	19	0.79
P34	Female	49	Right	11	> 10 years	5	Yes	1	19	0.79
P35	Female	18	Left	12	< 1 year	5	Not answered	2	20	0.80
P36	Male	24	Left	5	1–5 years	3	Yes	2	19	0.84
P37	Female	43	Left	15	> 10 years	3	Yes	3	11	0.91
P38	Female	25	Right	24	< 1 year	4	Yes	2	17	0.94

B Stimuli

Table B.1 describes the 40 items in the order of their identifiers. Genre and slant are the codes used in the analysis (Section 3.2). An item’s slant is “favors Left” when its central claim attacks a named politician or party of the right or defends one of the left, and “favors Right” in the mirror case; a dash means no identifiable slant. Items marked with a dagger (†) are the three false items whose label is debatable because they are misleading rather than fabricated: a list of promises that mixes real and false entries (LC26) and two propaganda videos that recontextualize real speech (LC29, LC39); Appendix F repeats the main analyses without them. “Week/set” gives the week and item set in which the item was sent; each set was delivered to the two schedules containing it, so every item was sent to 19 participants.

Table B.1. The 40 items with veracity, modality, genre, partisan slant, delivery schedule, and a short description.

ID	Truth	Modality	Genre	Slant	Week/set	Description
LC1	True	Image	News	–	1/A	Screenshot of a social media post claiming Colombia has the most road tolls in Latin America.

ID	Truth	Modality	Genre	Slant	Week/set	Description
LC2	False	Image	Warning	–	1/A	Screenshot of a WhatsApp status showing four men allegedly kidnapping children from schools and homes.
LC3	False	Image	Conspiracy	–	1/A	Infographic about “the real 2030 agenda” rewriting each sustainable development goal (“reducing population”, “abortion and contraception”).
LC4	False	Video	Health	–	1/A	A doctor claims dengue fever is caused by calcium deficiency and should be treated with calcium and vitamin D.
LC5	True	Text	Informative	–	1/A	Chain message on the harm of hummingbird feeders, recommending flowers instead.
LC6	True	Image	News	–	1/A	Table of Colombia’s minimum wage components for 2025.
LC7	True	Video	News	–	1/A	News report on new traffic barriers to be used by the mobility department of Cali.
LC8	False	Image	Political	favours Left	1/A	Screenshot of a fabricated tweet in which former president Iván Duque acknowledges a drug trafficker as a campaign financier.
LC9	True	Text+image	News	–	1/A	News article on the capture of a military-linked gang that supported illegal armed groups.
LC10	False	Text	Warning	–	1/A	Message about a new phone scam in which answering a call or text duplicates the victim’s SIM card.
LC11	True	Image	Informative	–	1/B	Infographic on what to do during a flood when in a vehicle.
LC12	False	Image	Health	–	1/B	Infographic with fabricated counts of deaths and adverse effects from COVID-19 vaccines, claiming the disease is poisoning rather than a virus.
LC13	False	Image	Warning	–	1/B	Screenshot of a WhatsApp status with information “direct from prison” warning of an order to steal children.
LC14	False	Video	Warning	–	1/B	A driver at a toll booth claims that “Law 769 of 2002” exempts drivers who wait five minutes from paying; no such provision exists, and he is let through.
LC15	True	Text	Health	–	1/B	Chain message framing exercise and fruit and vegetables as “medicine”.
LC16	True	Video	News	–	1/B	Official video about an associative credit line offered by a government agency.
LC17	False	Image	Political	favours Left	1/B	Bar chart with false figures for on-duty deaths of security forces across three presidential terms.
LC18	False	Image	Warning	–	1/B	Post claiming that pressing “cancel” twice at an ATM prevents data theft.
LC19	True	Video	Health	–	1/B	A doctor explains how to recognize the symptoms of a stroke.
LC20	True	Text+image	Informative	–	1/B	Message on the ecological benefits of opossums.
LC21	True	Image	Informative	–	2/C	Image of Article 199 of the Colombian Constitution.

	ID	Truth	Modality	Genre	Slant	Week/set	Description
1197	LC22	False	Video	Warning	–	2/C	Video of a young man putting on a mask, captioned as the masks Venezuelan immigrants use to rob.
1198							
1199	LC23	True	Video	News	–	2/C	An influencer describes a government technology training program.
1200							
1201	LC24	False	Video	Political	favours Left	2/C	Montage of fabricated or unverifiable anti-right images over a speech by Frédéric Bastiat read by an unknown narrator.
1202							
1203	LC25	True	Image	Informative	–	2/C	Post recommending cutting plastic rings before disposal to protect animals.
1204	LC26 [†]	False	Image	Political	favours Right	2/C	List of promises allegedly broken by President Gustavo Petro; the first four are real, the rest false or unverifiable.
1205							
1206	LC27	False	Text	Health	–	2/C	Long post claiming COVID-19 is a bacterium and is caused by blood coagulation.
1207							
1208	LC28	True	Video	News	–	2/C	Video of a Health Superintendency inspection of a drug dispensary in Bucaramanga.
1209	LC29 [†]	False	Video	Political	–	2/C	Propaganda video laying an anti-corruption speech by El Salvador’s president Nayib Bukele over images of former president Álvaro Uribe and the Colombian right, implying the remarks target them.
1210							
1211	LC30	True	Image	News	favours Right	2/C	Screenshot of a mainstream headline reporting that President Petro authorized security escorts for members of armed groups in the context of peace talks.
1212							
1213	LC31	False	Image	Warning	–	2/D	Photo of a fake letter attributed to a security company warning of robbers posing as Ministry of the Interior employees.
1214							
1215	LC32	True	Video	News	–	2/D	A pro-government influencer reports on a railway project of the Petro government.
1216	LC33	False	Image	Political	–	2/D	Screenshot of a post denouncing a corruption scandal over seized assets; conflates cases and contains false figures.
1217							
1218	LC34	True	Video	News	–	2/D	Excerpt from a national newscast on the capture of a criminal in Valle del Cauca.
1219	LC35	False	Image	Political	favours Left	2/D	Screenshot of a fabricated quote in which Congresswoman María Fernanda Cabal threatens a teachers’ union with a massacre.
1220							
1221	LC36	True	Image	News	–	2/D	Screenshot of a post stating that The Economist ranked Colombia among the best-performing economies of 2024 (sixth in the ranking shown).
1222							
1223	LC37	False	Text	Warning	–	2/D	Chain warning not to answer calls from certain numbers because foreign criminals can copy one’s contacts in three seconds.
1224							
1225							
1226							
1227							
1228							
1229							
1230							
1231							
1232							
1233							
1234							
1235							
1236							
1237							
1238							
1239							
1240							
1241							
1242							
1243							
1244							
1245							
1246							
1247							
1248							

Manuscript submitted to ACM

ID	Truth	Modality	Genre	Slant	Week/set	Description
LC38	True	Video	News	–	2/D	Official report on a municipal meeting with merchants about noise and use of public space.
LC39 [†]	False	Video	Political	favours Left	2/D	Propaganda video claiming the mainstream media are the true opposition to the Petro government.
LC40	True	Video	Informative	–	2/D	Video on precautions for motorcyclists at intersections with trucks.

C Summary of Quantitative Results

Table C.1 collects, for each result reported in Section 4.1 to Section 4.5, the unit of analysis, the estimate, and the test statistic. Where a result is reported at more than one unit of analysis in the text, the table gives the primary one; the judgment-level models behind the odds ratios are in Appendix D, and the same quantities recomputed under alternative analysis choices are in Appendix F.

Table C.1. Summary of quantitative results. CI = 95% confidence interval; GEE = generalized estimating equations clustered on participant; OR = odds ratio.

Result	Unit (<i>N</i>)	Estimate	Test
True items accepted vs. false items rejected	participant (38)	0.79 vs. 0.56	paired $t(37) = 5.20, p < .001, d_z = 0.84$
Response criterion <i>c</i> (bias toward “true”)	participant (38)	−0.30, CI [−0.42, −0.18]	$t(37) = -4.76, p < .001$; 29/38 negative
Sensitivity d'	participant (38)	0.90 (SD 0.76)	$t(37) = 7.34, p < .001$
Accuracy on true vs. false items	item (40)	0.78 vs. 0.57	Welch $t(37) = 3.60, p < .001$
Correct judgment, true vs. false item	judgment (662)	OR 2.78	GEE $z = 5.03, p < .001$
False videos vs. other false items detected	item (6 vs. 14)	0.41 vs. 0.63	Welch $t = -2.33, p = .049$; GEE interaction $z = 5.03$
Congruent vs. incongruent false political items believed	judgment (86)	0.51 vs. 0.19	GEE OR 4.47, $p = .009$
Majority accuracy, all 38 participants	item (40)	0.79, CI [0.64, 0.89]	Wilson interval over items; bootstrap over participants [0.68, 0.85]
Gain from accuracy weights estimated on held-out items	item split (1,500)	+0.006	$P(\text{gain} > 0) = 0.42$
Reliability of individual accuracy across item halves	participant (38)	$r = 0.23$	95% of splits $r > 0$
Replication of item verdicts across the two rater groups	item (40)	$r = 0.69$	$p < 10^{-6}$; majority verdicts agree on 32/40
Next answer “true” after disagreeing with the verdict	item pair (592)	OR 1.74	GEE $p = .001$
Next answer matches the next item’s majority, after disagreeing	item pair (548)	0.74 vs. 0.74	GEE $p = .47$
Next answer correct after disagreeing with the verdict	item pair (592)	coefficient 0.10	GEE $p = .60$

D Judgment-Level and Participant-Level Models

Table D.1 reports the judgment-level logistic models referred to in Section 3.6. Each model is estimated three ways: by GEE with an exchangeable correlation structure clustered on participants (the specification reported in the text), by GEE clustered on items (or on the next item, for the feedback models), and as a logistic regression with crossed random intercepts for participant and item fit by variational Bayes. Clustering on items widens the standard errors of item-level effects, as expected with 40 items, but every effect reported in the text keeps its sign and remains distinguishable from zero. Table D.2 reports the participant-level regression of accuracy on demographic covariates referred to in Section 4.1; no covariate predicts accuracy.

Table D.1. Judgment-level models. Coefficients are log-odds. GEE (part.): clustered on participant, standard errors in parentheses. GEE (item): clustered on item (on the next item for the last two models). Mixed: crossed random intercepts for participant and item, posterior standard deviations in parentheses.

Model	Term	GEE (part.)	z	GEE (item)	z	Mixed	z
correct ~ item is true	item is true	1.02 (0.20)	5.03	1.00 (0.28)	3.59	1.18 (0.14)	8.27
rated true ~ item is true	intercept	-0.23 (0.15)	-1.53	-0.25 (0.18)	-1.43	-0.26 (0.09)	-2.84
	item is true	1.50 (0.19)	7.91	1.51 (0.28)	5.41	1.75 (0.14)	12.30
correct ~ true × video	video	-0.93 (0.21)	-4.37	-0.92 (0.36)	-2.53	-0.94 (0.16)	-5.83
	item is true	0.44 (0.25)	1.79	0.44 (0.30)	1.46	0.56 (0.14)	3.96
	true × video	1.68 (0.33)	5.03	1.64 (0.58)	2.83	1.74 (0.24)	7.33
rated true ~ congruent (86 judgments)	congruent	1.50 (0.57)	2.61	1.57 (0.39)	4.02	1.77 (0.38)	4.67
next rated true ~ minority + own answer + next item true	minority	0.56 (0.17)	3.23	0.50 (0.19)	2.68	0.74 (0.21)	3.52
next correct ~ minority + next item true	minority	0.10 (0.19)	0.52	0.07 (0.20)	0.32	0.25 (0.20)	1.24

Table D.2. Participant-level ordinary least squares regression of accuracy on covariates ($n = 36$ participants with pre-survey data; $R^2 = 0.10$, $F(5, 30) = 0.64$, $p = .67$). Gender is coded 1 for men; political self-placement is a 1 (left) to 10 (right) scale; CRT is the number of correct cognitive reflection items.

Covariate	Coefficient	SE	<i>t</i>	<i>p</i>
Intercept	0.668	0.136	4.91	< .001
Age (years)	-0.0014	0.0017	-0.84	.41
Gender (man)	-0.067	0.054	-1.26	.22
Education (years)	0.0040	0.0052	0.77	.45
Political self-placement (1–10)	0.0023	0.0101	0.23	.82
CRT (0–3)	0.0044	0.0220	0.20	.84

E Crowd Composition Simulations

Table E.1 gives the simulated majority accuracy behind Figure 5b,c at three crowd sizes. Crowds were drawn without replacement from the 17 Left-identifying participants, from the 14 Right-identifying participants, half from each, or from all participants at random (1,500 crowds per cell); the majority verdict on each item was computed from the crowd members who had judged it. “Partisan” is the seven items with an identifiable slant (six false, one true), “other” the remaining 33. Averaged over all 40 items the four compositions differ by a few points at most; the differences are concentrated in the partisan items, and their direction follows the slant of the item set, since five of the six slanted falsehoods favored the Left. Because the Left-identifying participants were younger on average than the Right-identifying participants (31 versus 41 years; 11 of the 14 Right-identifying participants were women), the simulation cannot separate leaning from age or gender.

Table E.1. Simulated majority accuracy by crowd composition (1,500 crowds per cell). “Partisan” is the seven items with an identifiable slant; “other” is the remaining 33.

Crowd	Size 4			Size 8			Size 14		
	all	partisan	other	all	partisan	other	all	partisan	other
All Left	0.70	0.54	0.73	0.73	0.53	0.77	0.77	0.55	0.82
All Right	0.69	0.78	0.67	0.71	0.89	0.68	0.73	1.00	0.67
Balanced	0.70	0.68	0.71	0.74	0.72	0.74	0.78	0.77	0.78
Random	0.68	0.63	0.69	0.71	0.67	0.72	0.74	0.70	0.75

F Robustness Checks

Table F.1 recomputes the main results under four alternative analysis choices. (a) Without the three false items whose label is debatable (LC26, LC29, LC39; Section 3.2), the truth bias, the congruence effect, and the feedback effect are unchanged, but the video effect rests on four false videos and is no longer significant. (b) Coding the two ambiguous political items (LC29, LC33) as favoring the Left, as the paper’s slant rule could be read to require, adds two items and 29 judgments to the congruence analysis and leaves the effect intact. (c) Clustering the judgment-level models on items rather than participants (full estimates in Table D.1) widens standard errors but changes no conclusion. (d) Item verdicts replicate across the two disjoint groups of raters who judged each item far better than individual accuracy replicates across halves of the items, which is the basis for the claim in Section 4.4 that items were well measured and raters were not. The last row group supports the design implication in Section 6.1 that interfaces should show how contested a verdict is rather than a bare label: items on which the community split narrowly were wrong far more often than items with a lopsided vote.

G Interview Codebook Summary

Table G.1 lists the 30 codes from the reflexive thematic analysis of the nine interviews (Section 3.6), grouped into the four themes reported in Section 5, with the number of interviewees whose account instantiates each code. The full codebook, with code definitions, participant-level assignments, and quotations in Spanish with English translations, is provided in the supplementary material together with the analysis code and the pseudonymized judgment data.

Table F.1. Robustness checks. Row groups correspond to the alternative analysis choices described in the text; “main” repeats the estimate reported in the paper for comparison.

Quantity	Main analysis	Alternative
<i>(a) Without LC26, LC29, LC39 (608 judgments)</i>		
Hit rate / false-alarm rate	0.79 / 0.45	0.79 / 0.43
Sensitivity d' / criterion c	0.90 / -0.30 (29/38 negative)	0.95 / -0.28 (28/38 negative)
False videos vs. other false items detected	0.41 (6) vs. 0.63 (14), $p = .049$	0.47 (4) vs. 0.62 (13), $p = .22$
Congruent vs. incongruent falsehoods believed	0.51 vs. 0.19, OR 4.47, $p = .009$	0.48 vs. 0.19, OR 4.19, $p = .042$
Next answer “true” after disagreement, OR	1.74, $p = .001$	1.70, $p = .004$
<i>(b) Alternative slant coding: LC29 and LC33 favor the Left (8 items, 115 judgments)</i>		
Congruent vs. incongruent falsehoods believed	0.51 (39) vs. 0.19 (47)	0.58 (55) vs. 0.27 (60), OR 3.74, $p = .005$
<i>(c) GEE clustered on items instead of participants</i>		
Correct judgment, true vs. false item	$z = 5.03$	$z = 3.59$, $p < .001$
True $\times$ video interaction	$z = 5.03$	$z = 2.83$, $p = .005$
Congruence, OR	4.47, $z = 2.61$	4.81, $z = 4.02$
Next answer “true” after disagreement, OR	1.74, $z = 3.23$	1.65, $z = 2.68$
Next answer correct after disagreement	$z = 0.52$	$z = 0.32$
<i>(d) What replicates: items vs. raters</i>		
Share judging an item true, rater group A vs. group B (40 items)	—	$r = 0.69$, $p < 10^{-6}$; majorities agree on 32/40
Congruent side believes the item more, per item	—	6 of 6 items (sign test $p = .031$)
Participant accuracy, one half of items vs. the other	$r = 0.23$	95% of splits $r > 0$
<i>(e) Majority verdict correctness by vote margin</i>		
Margin below 60/40 (11 items)	—	68% correct
Margin 60/40 to 75/25 (12 items)	—	75% correct
Margin above 75/25 (17 items)	—	88% correct

Table G.1. Interview themes and codes, with the number of interviewees (of nine) whose account instantiates each code.

Code	<i>n</i>
<i>How participants decided</i>	
Prior knowledge and lived experience	6
External search (Google, news); once only for two	6
Social consultation or debate before answering	5
Political stance; propaganda treated as false	4
Videos deferred to a quiet moment; texts answered anywhere	4
Surface and production cues	3
Shortcuts: skimming, balancing answers, guessing when media failed	3
Staying inside WhatsApp so as not to lose the task	2
Local social-evidence test for alarms	1
Unarticulated intuition	1
<i>Reactions to the community verdict</i>	
Agreement read as confirmation	6
Disagreement prompted self-doubt or re-reading	5
Disagreement attributed to the community's knowledge, age, or politics	4
Verdict given little attention	3
Trust in the aggregate conditional on who is in it	3
Verdict experienced as a game	2
Meaning of the percentages unclear	1
<i>Motivation</i>	
Fitting the task into daily routines	9
Payment	6
Learning, curiosity, interest	6
Local relevance and realism of the content	5
Pre-existing critical stance toward forwarded content	3
Study prompted more checking than usual	2
Commitment to the researcher	1
<i>Conditions for a real system</i>	
Cadence of one or two items a day, with deadlines	6
WhatsApp channel easy to use	5
Purpose and tangible action	3
Engagement design (attention, room for opinions)	3
Trust and scam fear as the entry barrier	2
Representativeness as a condition of legitimacy	2
WhatsApp lacks a community-warning layer	1
Technical failure	1